

CACM — Continuous Attention Coordination Module

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Category: AI & Interpretation

Subcategory: Continuous Attention Coordination

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Authority: OOF

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Canonical Language: English

Canonical Definition

Continuous Attention Coordination Module defines the structural conditions under which a continuously present intelligent system may detect, distribute, synchronize, prioritize, shift, and preserve attention across simultaneous interaction streams, modalities, contexts, and live operational signals without collapsing into unstable focus, hidden prioritization, uncontrolled switching, or behaviorally opaque coordination logic.

A system satisfies CACM only if:

- simultaneous attention handling remains explicitly governed
- active focus states are structurally identifiable
- switching between streams remains bounded and reviewable
- prioritization logic does not silently distort interaction conditions
- continuous attention remains coordinated without losing runtime interpretability

A system that manages multiple interaction streams without explicit focus logic, priority conditions, or interpretable coordination states does not satisfy CACM.

Module Function

CACM defines the attention coordination layer of continuous interaction governance.

It ensures that a system operating across live multimodal and multi-stream environments does not treat attention as an invisible internal convenience, but as a governed operational condition affecting interaction quality, timing, intervention, and behavioral validity.

The module applies wherever a system must coordinate attention across:

- voice and text simultaneously
- multimodal sensor streams
- multiple human participants
- environmental signals and direct interaction
- proactive alerts and ongoing dialogue
- parallel workflows
- live orchestration environments
- continuous contextual updates

Its function is not merely to allow multi-stream processing.

Its function is to ensure that multi-stream processing remains governable as attention architecture.

Step 1 — Define the Attention Object

The organization must define what exactly the system may attend to during continuous participation.

Minimum requirement:

- the attention object is explicit
- the scope of attention coordination is structurally bounded
- undefined attention targets are excluded from valid governance logic

The attention object may include:

- speakers
 - text streams
 - visual inputs
 - environmental signals
 - active tasks
 - safety alerts
 - system coordination channels
 - context-preserving interaction states
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Step 2 — Define Attention States

The system must define which attention states may exist during runtime interaction.

Minimum requirement:

- attention states are explicit
 - the difference between passive availability, active focus, background monitoring, divided attention, and escalated
 - priority is identifiable
 - attention does not remain behaviorally active through undefined internal states
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Attention states may include:

- idle
- passive monitoring
- active focus
- split attention
- background attention
- escalated priority focus
- redirected focus
- suspended focus

Step 3 — Define Prioritization Logic

The system must define how attention priority is assigned when multiple streams compete.

Minimum requirement:

- prioritization logic is explicit
- critical and non-critical streams remain distinguishable
- attention priority does not depend on opaque internal preference alone

Priority conditions may include:

- safety relevance
- active human interaction
- task continuity
- escalation state
- interruption necessity
- environmental urgency
- system coordination dependency
- runtime trust conditions

Without prioritization logic, continuous interaction becomes behaviorally unstable under multi-stream load.

Step 4 — Define Attention Switching Conditions

The system must define when attention may shift from one stream, participant, context, or modality to another.

Minimum requirement:

- switching conditions are explicit
- attention does not jump unpredictably across streams
- focus change remains governed by valid runtime conditions

This means the system must remain able to determine:

- when attention stays
- when attention shifts
- when attention splits

- when attention returns
- when attention escalation overrides current focus

Step 5 — Define Coordination Boundaries

The system must define the limits of continuous attention coordination.

Minimum requirement:

- coordination boundaries are explicit
- simultaneous streams do not silently expand into unlimited monitoring scope
- attention handling remains bounded by system capacity, governance conditions, and interaction validity
- requirements
- This prevents a system from appearing coordinated while actually operating under uncontrolled attention spread.

Step 6 — Preserve Attention Traceability

The system must preserve traceability of focus states, switching events, priority changes, and coordination outcomes.

Minimum requirement:

- attention-state changes are reviewable
- switching logic remains reconstructable
- later audit can determine what the system attended to, when it shifted, why it prioritized, and how that affected
- interaction behavior

If attention coordination cannot be reconstructed, then multi-stream interaction becomes behaviorally opaque.

Step 7 — Restrict Invalid Attention Design

The system must not be treated as valid if attention coordination remains hidden, unstable, structurally unbounded, or behaviorally uninterpretable during continuous operation.

Minimum requirement:

- invalid attention conditions are identifiable
- opaque prioritization is excluded
- uncontrolled attention switching is blocked, bounded, or invalidated where governance requires interpretable
- coordination

Scenario

A realtime assistant interacts with a user through voice while simultaneously processing text, environmental signals, and

task-relevant alerts inside a live environment.

Application

CACM is used to ensure:

- active focus and background monitoring remain distinguishable
 - priority across streams remains governed
 - switching between human dialogue and environmental alert conditions remains reviewable
 - multi-stream coordination does not collapse into hidden or unstable attention logic
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Result

The system gains a stronger continuous interaction architecture in which multimodal attention remains bounded, interpretable, and behaviorally governable.

Scenario

A continuously present intelligent system coordinates multiple participants, environmental signals, and live workflow conditions in a shared operational setting.

Application

CACM is used to ensure:

- simultaneous attention demands remain structured
 - focus escalation does not silently override valid interaction boundaries
 - multi-participant coordination remains reviewable
 - later reconstruction can determine how attention was allocated across competing streams
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Result

The environment gains a stronger live participation structure in which attention handling remains governed rather than becoming an opaque source of behavioral instability.

Canonical Closing Statement

If continuous attention cannot remain governed across simultaneous streams, presence becomes active but coordination becomes structurally unreliable.

Related Documents

→ Continuous Interaction Layer (CIL™)

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Canonical Source

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